

Aufgabe 1:

$$a) \frac{3}{7} + \frac{7}{3} = \frac{3 \cdot 3 + 7 \cdot 7}{3 \cdot 7} = \frac{9 + 49}{21} = \frac{58}{21}$$

$$b) \left(\frac{1}{3} + \frac{2}{9} \right) \cdot \left(\frac{11}{8} - \frac{1}{4} \right) = \left(\frac{3+2}{9} \right) \cdot \left(\frac{11-2}{8} \right) = \frac{5 \cdot 9}{9 \cdot 8} = \frac{5}{8}$$

$$c) \left(\frac{13}{2} \cdot \frac{169}{12} \right) : \frac{3}{26} = \frac{13}{2} \cdot \frac{12}{13 \cdot 13} \cdot \frac{26}{3} = \frac{13 \cdot 2 \cdot 26}{2 \cdot 13 \cdot 13 \cdot 3} = \frac{26}{13 \cdot 3} = \frac{2}{3}$$

$$d) 5^{-2} \cdot 100 = \frac{100}{5^2} = \frac{100}{25} = \frac{4 \cdot 25}{25} = 4$$

$$e) -2(2x-4) + x(1+x) \cdot (-1) = -4x + 8 - (x + x^2) = -5x + 8 - x^2$$

$$f) -121ab^3 - (11a^2b)^2 \cdot (-2a^{-3}b) \\ = -121ab^3 + (121a^4b^2 \cdot 2a^{-3}b) \\ = -121ab^3 + 242ab^3 = 121ab^3$$

$$g) -\frac{a+b}{c} - \frac{a-b}{c} = \frac{-a-b-a+b}{c} = \frac{-2a}{c}$$

Aufgabe 2

$$b(a-b) + b-a \\ = b(a-b) - (a-b) = (a-b)(b-1)$$

$$a) -1 \Rightarrow -(1-b)(a-b)$$

$$b) a-b \Rightarrow (a-b)(b-1)$$

$$c) b-a \Rightarrow (b-a)(1-b)$$

Aufgabe 3

$$a) \frac{a-b}{a-b} = 1 \quad b) \frac{a-b}{b-a} = \frac{a-b}{-(a-b)} = -1 \quad c) \frac{a^2+b^2}{a+b} \text{ (nicht mehr)}$$

$$d) \frac{a^2-b^2}{a-b} = \frac{(a+b)(a-b)}{a-b} = a+b \quad e) \frac{2a+3b}{2c+3d} \text{ (nicht mehr)}$$

$$f) \frac{k^2+k^3}{k^2} = \frac{k^2(1+k)}{k^2} = 1+k \quad g) \frac{2k^2}{4k^2+6k^3} = \frac{2k^2}{2k^2(2+3k)} = \frac{1}{2+3k}$$

K²

n

in + on

an (2+3K)

2 3 K

h) $\frac{1+K^2}{1+K^3}$ (nicht mehr)

Aufgabe 4

c) $\sqrt{2} \cdot \sqrt{32} = \sqrt{2 \cdot 32} = \sqrt{64} = \sqrt{8^2} = 8$

b) $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$

Nein: Bz. $\begin{matrix} a=1 \\ b=4 \end{matrix} \Rightarrow \sqrt{1+4} = 3 \neq \sqrt{1} + \sqrt{4} = 1+2=3$

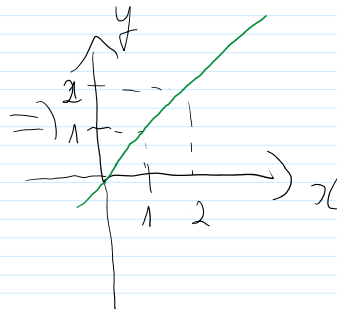
Au $\frac{1}{\sqrt{5}} = \frac{1}{\sqrt{5}} \cdot \frac{\sqrt{5}}{\sqrt{5}} = \frac{\sqrt{5}}{5}$

$$\frac{1}{3+\sqrt{5}} = \frac{\sqrt{3}-\sqrt{5}}{(\sqrt{3}+\sqrt{5})(\sqrt{3}-\sqrt{5})} \quad (\text{Theory } (a-b)(a+b) = a^2 - b^2)$$
$$= \frac{\sqrt{3}-\sqrt{5}}{3-5} = \frac{\sqrt{3}-\sqrt{5}}{-2} = \frac{\sqrt{5}-\sqrt{3}}{2}$$

Aufgabe 5

a) $y = x$

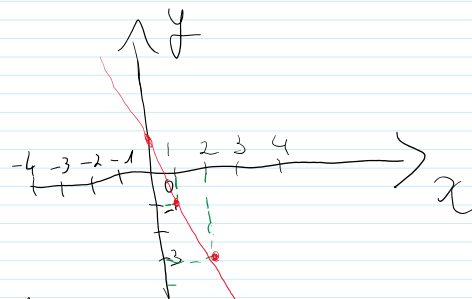
• A(1,1), B(2,2)



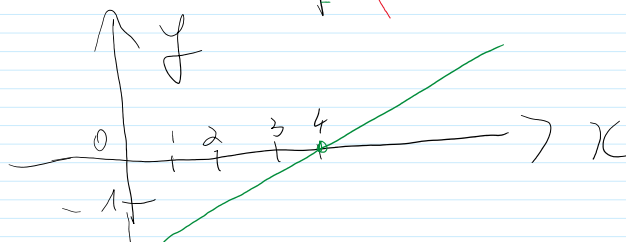
b) $y = -2x + 1$

A: $x=1 \Rightarrow y=-1$

: $x=2 \Rightarrow y=-3$



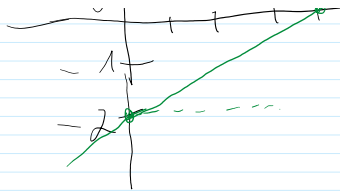
c) $y = \frac{1}{2}x - 2$



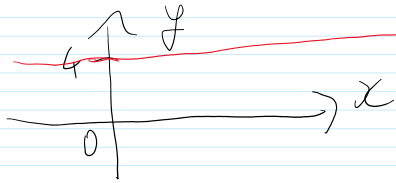
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A: $x=0; y=-2$

B: $y=0; x=4$



d) $y=4$



e)
$$\begin{cases} y = -2x + 1 \\ y = \frac{1}{2}x - 2 \end{cases} \Rightarrow \begin{cases} -2x + 1 = \frac{1}{2}x - 2 \\ \Rightarrow x = \frac{6}{5} \Rightarrow y = -\frac{7}{5} \end{cases}$$

Aufgabe 6

a) $y = ax + b$: $\begin{cases} A: x=8; y=0 \\ B: x=0; y=2 \end{cases} \Rightarrow \begin{cases} 8a + b = 0 \\ b = 2 \end{cases} \Rightarrow \begin{cases} a = -\frac{1}{4} \\ b = 2 \end{cases}$

$\Rightarrow y = -\frac{1}{4}x + 2$

b) $\begin{cases} A: x=0; y=0 \\ B: x=2; y=-1 \end{cases} \Rightarrow \begin{cases} b=0 \\ 2a+b=1 \end{cases} \Rightarrow \begin{cases} a = -\frac{1}{2} \\ b = 0 \end{cases}$

$y = -\frac{1}{2}x$